Module 4: Cancer

Team Members: Will Marschall and Dani Folks

Project Title: Do proliferative-signaling and apoptosis-evasion expression programs track tumor stage in colon adenocarcinoma?

Project Goal:

Our project seeks to determine whether two cancer hallmarks (**sustained proliferative signaling** and **resisting apoptosis**) show coordinated RNA-expression patterns in **colon adenocarcinoma (COAD)** and whether those patterns may relate to tumor stage.

Our current working hypothesis is that **COAD tumors with higher expression of proliferative-signaling genes will also show stronger apoptosis-resistance patterns and will tend to fall into more advanced tumor-stage groups**, although this remains a hypothesis to be tested over Module 4 rather than a conclusion.

Disease Background:

Cancer hallmark focus

For this project, we decided to focus in on two hallmarks that are mechanistically connected and also easy to operationalize with our RNA-expression data:

1. **Sustained growth/proliferative signaling**
   In our quick reference handout, we saw that this hallmark is described as cancer cells no longer waiting for normal external growth signals before dividing. This hallmark increases the cancer-cell division rate and relates idea to overactive growth-factor receptors and downstream signaling genes such as **EGFR, RAS/KRAS, BRAF, and MYC** [1]. In healthy tissue, proliferation is tightly regulated by extracellular ligands (e.g., growth factors) binding to receptors, which activate intracellular signaling cascades only when appropriate. In cancer, this control is bypassed through multiple mechanisms that lead to constitutive activation of growth pathways, including:

   - Overexpression or mutation of receptor tyrosine kinases (e.g., EGFR, ERBB2), allowing signaling even in the absence of ligand
   - Activating mutations in downstream signaling proteins (e.g., KRAS, NRAS, BRAF), which lock pathways such as the MAPK/ERK cascade in an “on” state

- Activation of survival and growth pathways such as PI3K/AKT signaling, promoting both proliferation and metabolic support
- Upregulation of transcription factors (e.g., MYC, FOS, JUN) that directly increase expression of genes required for cell-cycle progression

2. **Resist apoptosis**
   This hallmark describes the ability of cancer cells to evade/resist programmed cell death, a tightly regulated process that normally removes damaged or abnormal cells. In healthy tissue, apoptosis is particularly triggered by DNA damage, oncogenic stress, or loss of survival signals, and is executed through coordinated activation of caspases via intrinsic (mitochondrial) and extrinsic (death-receptor) pathways. For this project, our apoptosis-resistance genes includes **TP53, BCL2, BCL2L1, MCL1, CASP8, CASP3, BAX, BAK1, APAF1, BID, CASP9, FAS, FASLG, BAD, BBC3, BCL2L11, CFLAR, DIABLO, TNF, and TNFRSF10B.** In cancer, this system is disrupted through multiple mechanisms which include:

   - Upregulation of anti-apoptotic proteins **(e.g., BCL2, BCL2L1, MCL1)**, which stabilize mitochondrial membranes and prevent cytochrome c release
   - Downregulation or mutation of pro-apoptotic factors **(e.g., BAX, BAK1, BID)**
   - Loss of tumor suppressor function, especially **TP53**, which we assume would normally promote apoptosis in response to DNA damage
   - Impaired caspase activation, limiting execution of the apoptotic program

   In a functional sense, resisting apoptosis allows cancer cells to survive despite accumulating mutations as well as sustain growth under stress (and during therapy).

Why we chose colon adenocarcinoma (COAD)

We narrowed our analysis to **one cancer type** instead of many because we were contemplated on deciding whether to focus on one cancer type or to compare multiple cancers, and a single-type analysis is less biologically heterogeneous for our machine-learning workflow discussed in depth during class. We chose **colon adenocarcinoma (COAD)** because:

- Colorectal cancer was emphasized in our 1st lecture and hallmark simulation discussion
- Colon cancer has well-established links to deregulated growth signaling pathways
- Stage information is clinically meaningful for this disease in our initial examination
- One cancer type is a cleaner starting point than attempting many tumor biologies on an initial workflow.
- Colorectal cancer is one of the most common cancers worldwide, ranking third in incidence globally and contributing substantially to cancer-related mortality. [5] This provides us with a strong ethical and scientific motivation to further investigate COAD due to the large number of individuals it affects.

Overview of hallmark biology

In normal tissues, cells only divide when instructed by extracellular signals, undergo apoptosis when damaged or no longer needed, and remain integrated within the structural and functional organization of the tissue. These processes are largely controlled through coordinated signaling networks that maintain tissue homeostasis and also balance cell proliferation with differentiation and death. Cancer can be easily understood as the breakdown of these regulatory systems, where cells progressively acquire the ability to override multiple biological constraints at the same time. Rather than a single mutation causing disease, tumor development illustrates how the accumulation of changes disrupting normal control mechanisms can allow cells to behave in an erratic manner.

For this project, we focus on two complementary hallmarks: one that acts as the “gas pedal” by driving sustained cell division, and one that weakens the “brake pedal” by impairing programmed cell death, the same analogy used in class to distinguish between oncogenes and tumor suppressors. Together, these hallmarks help us define a two-axis framework in which tumors can be positioned based on how strongly they promote growth versus how effectively they evade death. This paired analysis provides a biologically interpretable way to assess tumor behavior using our provided RNA-expression data.

Genes associated with the hallmark to be studied

We expanded our hallmark panel to 40 genes total so that the model has a richer feature space while remaining biologically interpretable.

Sustained proliferative signaling genes `EGFR, ERBB2, KRAS, NRAS, BRAF, PIK3CA, AKT1, MYC, FOS, JUN, MET, IGF1R, FGFR1, FGFR2, EGF, HGF, SOS1, GRB2, MAPK1, MAPK3`

These genes span multiple levels of growth signaling pathways, including:

- Receptor-level activation (EGFR, ERBB2)
- Intracellular signal transduction (KRAS, NRAS, BRAF, PIK3CA, AKT1)
- Transcriptional regulation of the cell cycle (MYC, FOS, JUN)

Resist apoptosis genes `TP53, BCL2, BCL2L1, MCL1, CASP8, CASP3, BAX, BAK1, APAF1, BID, CASP9, FAS, FASLG, BAD, BBC3, BCL2L11, CFLAR, DIABLO, TNF, TNFRSF10B`

These genes represent key regulators of apoptotic pathways, including:

- Tumor suppressor signaling (TP53)
- Mitochondrial apoptosis regulation (BCL2 family: BCL2, BCL2L1, MCL1, BAX, BAK1, BID)
- Execution of apoptosis via caspases (CASP8, CASP3)

- Apoptosome formation and signaling (APAF1)

A CSV version of our list is included in our repo as `hallmark_gene_sets.csv`. In the EDA step, we explicitly check which of these genes are present in the local training matrix before making any downstream claims.

Prevalence & incidence

Colorectal cancer is a major public-health burden. The American Cancer Society estimates that in the United States in **2026** there will be about **108,860 new colon cancer cases** and **49,990 new rectal cancer cases**, and colorectal cancer is expected to cause about **55,230 deaths**. Excluding skin cancers, it is the **third most common cancer** diagnosed in both men and women in the United States [5].

Risk factors (genetic, lifestyle) & societal determinants

Major risk factors for colorectal cancer include **older age**, **family history**, **personal history of colorectal neoplasia**, inherited syndromes such as familial adenomatous polyposis **FAP** and **Lynch syndrome**. Lifestyle-related factors also play a role, including **dietary patterns**, **alcohol consumption**, **cigarette smoking**, and **obesity** [4], [5]. All of which can largely influence inflammation and cellular stress within the colon. **Inflammatory bowel disease (IBD)** further increases risk by repeated cycles of tissue damage and regeneration, which can facilitate mutation accumulation.

In addition to biological risk factors, we note that societal determinants can significantly contribute to risk factors, including access to screening, healthcare availability, socioeconomic status, and geographic location (strongly influence both incidence and outcomes). Early detection can significantly reduce mortality by identifying precancerous lesions or early-stage tumors, whereas limited access to care is associated with later-stage diagnoses and poorer prognosis [5].

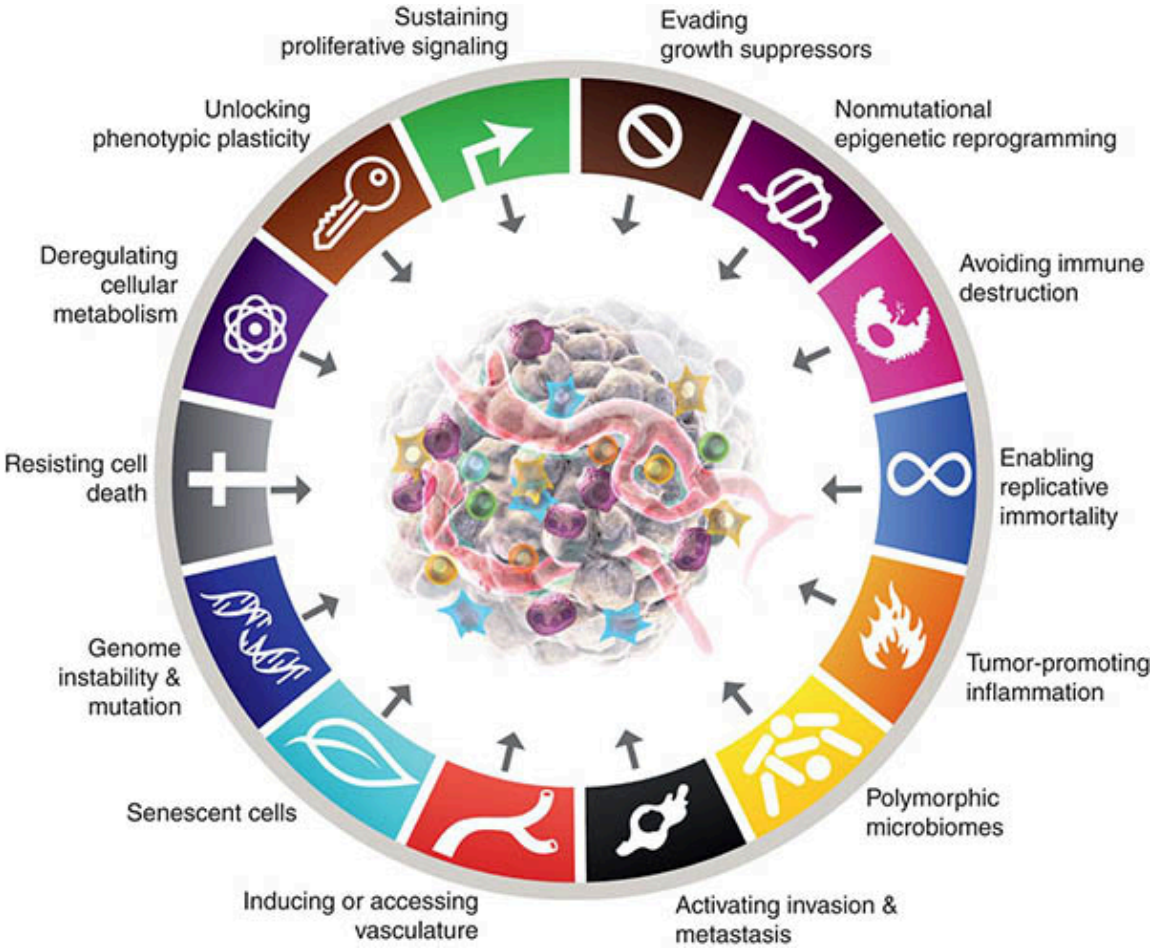
Standard of care treatments (& reimbursement)

Treatment depends strongly on stage and biomarker status. For early-stage disease, surgical resection is often the primary treatment and can be curative. Standard approaches include **surgery**, **chemotherapy**, **radiation therapy** (more common in rectal disease), **targeted therapy**, and **immunotherapy**.

While we do not make detailed reimbursement claims in this analysis, it is important to note that both tumor stage and biomarker status directly influence treatment selection as well as eligibility for targeted therapies and overall cost of care. This further supports our emphasis on the relevance of linking molecular patterns (such as hallmark-associated gene expression) to clinically meaningful variables like tumor stage.

Biological mechanisms (anatomy, organ physiology, cell & molecular physiology)

The colon absorbs water and electrolytes from intestinal contents, while the rectum stores stool before elimination. Colon adenocarcinoma arises from epithelial cells in this system. At the molecular level, colorectal tumor development is driven by changes in genes controlling cell division, survival, differentiation, and tissue organization. That makes COAD a good test case for our RNA-based hallmark analysis because aberrant growth signaling and resistance to cell death should both leave measurable transcriptomic signatures for us to interpret [4].



Data-Set:

The dataset for this project comes from **The Cancer Genome Atlas (TCGA) RNA-seq data reprocessed by Rahman et al.** and distributed through **GEO accession GSE62944** [2], [3]. Dr. Groves further subsetted those data to a smaller teaching dataset containing approximately **50–100 tumors per cancer type** across **24 cancer types** (about **1802 tumors total**) with a reduced metadata table. It should also be noted that not all metadata fields are populated equally across cancer types, so checking missingness is an explicit part of our EDA.

What these files contain

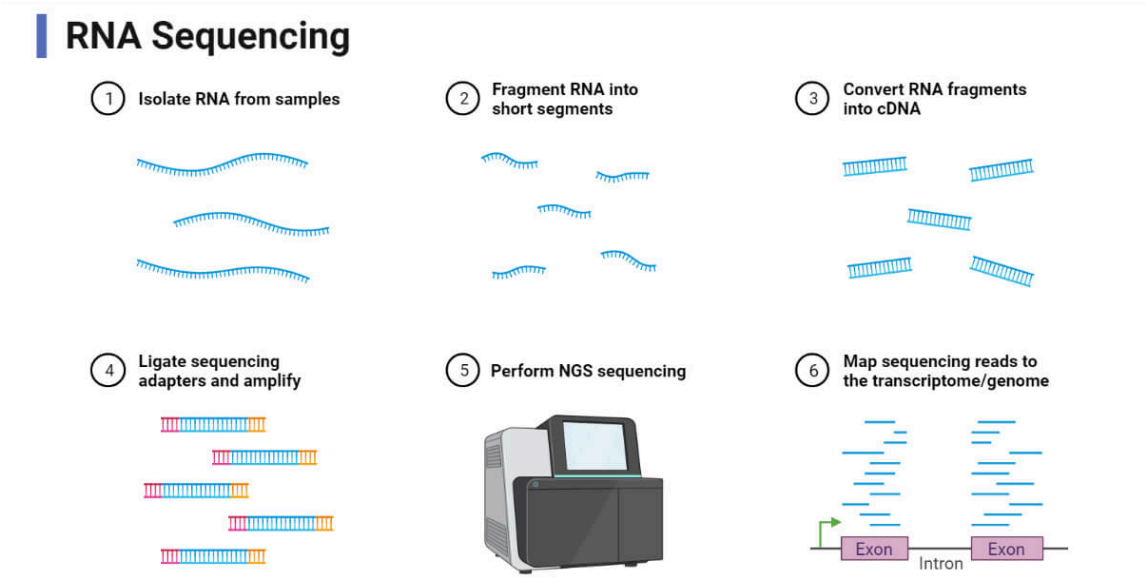
- `TRAINING_SET_GSE62944_subsample_log2TPM.csv`
  Gene-expression matrix with **genes as rows** and **tumor samples as columns**.
- `TRAINING_SET_GSE62944_metadata.csv`
  Metadata table with **samples as rows** and clinical/annotation variables as columns.
- `VALIDATION_SET_...` and `TEST_SET_...`
  We used the **validation set** to compare candidate models and choose our final model, and we used the **test set once at the end** to estimate final out-of-sample performance on completely unseen COAD samples.
- `GSE62944_metadata_percent_nonNA_by_cancer_type.csv`
  A provided summary of metadata completeness by cancer type.

What is RNA-seq and how should the values be interpreted?

RNA sequencing measures the abundance of RNA transcripts in a sample, which gives a snapshot of our **transcriptome**. In this dataset, the values are **log2(TPM + 1)**:

- **TPM** stands for **transcripts per million**, a normalization that accounts for gene length and sequencing depth.
- The **log2(TPM + 1)** transform compresses the dynamic range so highly expressed genes do not dominate as strongly and zero counts remain defined [2].

Because our dataset is already normalized and log-transformed, it is appropriate for exploratory plots, correlation analysis, and many machine-learning workflows. However, the expression values are **not raw counts**, so count-based methods like DESeq2 would require different input data.



How the data were generated

Rahman et al. describe reprocessing TCGA RNA-seq data using the **Rsubread** pipeline, aligning reads to the **hg19** reference genome, generating **gene-level integer counts** with `featureCounts`, and then converting those to normalized **FPKM** and **TPM** values. The GEO record states that the dataset contains **9264 tumor samples and 741 normal samples across 24 cancer types** in the full public compendium, along with matched clinical variables [3].

Metadata we expect to use

The most important of our metadata fields are:

- `cancer_type` to isolate **COAD**
- a **tumor stage** field, if available/populated for COAD in the training split
- possibly age/sex or other basic covariates if available and relevant later on in our analysis

Because stage-column names can differ across datasets, our EDA script searches several likely stage-column names automatically instead of assuming one fixed label.

What we do in the EDA

Our EDA script is designed to answer these questions:

- What is the size of the training matrix?
- What are the minimum, maximum, and average expression values?
- Which genes have the highest average expression?
- What metadata are missing?
- Are our hallmark genes actually in the dataset?
- How many COAD samples are in the training split?
- Can we visualize the two hallmark programs with scatterplots and PCA?

Data Analysis:

Methods

We utilized **unsupervised learning** methods to explore structure in the COAD samples, following the dimensionality-reduction and clustering workflow introduced to us in detail lecture. We first cleaned the metadata by converting age to numeric and simplifying AJCC tumor-stage labels into broader **Stage I**, **Stage II**, **Stage III**, and **Stage IV** bins. We also created a binary stage label (E**arly = Stage I/II**, **Late = Stage III/IV)** for our supervised-learning model.

We then subset the expression matrix to COAD and to a **40-gene hallmark panel**: **20 genes** associated with **sustained proliferative signaling** and **20 genes** associated with **resisting**

cell death. Our resulting sample-by-gene matrix was standardized before machine learning so that genes with different expression ranges would contribute more comparably to our downstream analyses.

We used **PCA** as our first dimensionality-reduction method because it provides a linear summary of the strongest axes of variation in the selected hallmark genes. We then used **K-means clustering** to look for exploratory subgroups in the standardized gene-expression matrix. K-means minimizes the within-cluster distance between samples and their assigned centroids, so it provides us with a great and simple first pass at identifying molecular groupings. We evaluated candidate values of k from **2 to 6** using silhouette score and selected the **best-performing k**. We also used **UMAP** as a nonlinear dimensionality-reduction method to compare whether local neighborhood structure looked similar to our PCA view.

For supervised learning, we used **logistic regression** to classify simplified tumor stage (**Early = Stage I/II**, **Late = Stage III/IV**) from hallmark-expression features in COAD. We chose logistic regression because our target is binary, our sample size is modest, and the model provides interpretable coefficients while also supporting regularization.

Following the workflow discussed in class, we used the **training dataset** to fit candidate models, the **validation dataset** to compare models and judge generalization, and the **test dataset only once at the end** to estimate final out-of-sample performance on unseen data. Our baseline model used the 40 hallmark genes alone. Our improved model added biologically meaningful summary features (**proliferative score**, **apoptosis-resistance score**, and **age at diagnosis**) and used class balancing plus stronger regularization to reduce overfitting.

To evaluate performance, we used both **threshold-dependent** and **threshold-independent** metrics. At the default threshold of 0.5, we reported **accuracy**, **balanced accuracy**, **precision**, **recall**, **F1 score**, and the **confusion matrix**. Across all thresholds, we reported **AUROC**. After selecting the better-performing model using validation performance, we applied that final model to the **held-out test set** and reported the final out-of-sample metrics.

Analysis

1. Set up the notebook and import our analysis script

In this first step, we set up the notebook environment and import our main.py script from the code folder. This lets us keep the full analysis logic in one reusable Python file while still showing the major workflow steps inside the notebook.

In [ ]:
```
# This is a short snippet from our main.py script.
# The full script is quite long, so we are only showing the top portion here where
# Please view our main.py script for full detailed comments, methodologies, and res
```

```
from pathlib import Path
import sys
import importlib
import pandas as pd
from IPython.display import display, Image

pd.options.display.float_format = "{:.6f}".format

NOTEBOOK_DIR = Path.cwd()
PROJECT_ROOT = NOTEBOOK_DIR.parent if NOTEBOOK_DIR.name.lower() == "code" else NOTE
CODE_DIR = PROJECT_ROOT / "code"

if str(CODE_DIR) not in sys.path:
    sys.path.append(str(CODE_DIR))

import main
importlib.reload(main)
```

2. Run the full exploratory workflow on the COAD training set

This cell runs the full exploratory workflow defined in `main.py`. It loads the training expression matrix and metadata, subsets the data to COAD, checks whether the selected hallmark genes are present, cleans the metadata, computes hallmark scores, performs PCA and K-means clustering, and generates the summary figures saved to the `results` folder.

In [ ]:
```
# We wrap the entire EDA workflow in a function and call it in the main block.
# This shows our entire EDA workflow in one place and allows us to easily reuse the
eda_outputs = main.run_eda()
```

3. Display the global training-set summary

These outputs summarize our overall training expression matrix before subsetting to COAD. The first table reports the total number of genes and samples as well as the global minimum, maximum, mean, and median log2(TPM+1) expression values. The second table lists the genes with the highest average expression across the entire training set, which helps us understand the overall scale and composition of the dataset before narrowing to our hallmark-focused analysis.

In [ ]:
```
# After running the EDA workflow, we can access all the major tables and results fr
display(eda_outputs["full_summary"])

# For example, we can display the top mean expression table which shows the top 10
display(eda_outputs["top_mean_expression"])
```

4. Examine metadata completeness and the cleaned COAD metadata

Because metadata completeness varies across cancer types in this teaching dataset, we explicitly examined missingness before moving forward. We also displayed the cleaned COAD metadata to confirm that age was converted to numeric where possible and that the

detailed AJCC labels were successfully simplified into broader stage groups. This cleaning step is important because it prepares a more interpretable stage variable for plotting now and for supervised learning later.

In [ ]:
```
# We can also display the missingness table which shows the percentage of missing v
# also the cleaned cancer metadata with the new stage columns we created.
display(eda_outputs["missingness"].head(20))

# Display the first 10 rows of the cleaned cancer metadata with the new stage colum
display(
    eda_outputs["cancer_metadata"][
        ["age_at_diagnosis", "stage_clean", "stage_simple", "stage_binary"]
    ].head(10)
)
```

5. Confirm that the selected hallmark genes are present in the COAD subset

Before interpreting any downstream result, we verified that the requested hallmark genes were actually present in the local expression matrix. The presence table confirms which genes were found, and the summary-statistics table provides the mean, median, and variance of each hallmark gene across the COAD samples. This step ensures that the selected features are available and gives a first look at which genes are relatively stable or variable within this cohort.

In [ ]:
```
# Finally, we can also display the hallmark gene presence table which shows the pre
# Also the summary statistics for these genes.
display(eda_outputs["hallmark_gene_presence"])
display(eda_outputs["hallmark_gene_stats"])
```

6. Review the sample-level hallmark-score table

We next computed one simple score per hallmark by averaging expression across the genes in each hallmark set for each sample. These sample-level hallmark scores give us a compact way to compare the two biological programs across tumors and provide a natural bridge between gene-level data and stage-based comparisons.

In [ ]:
```
# We can also display the merged scores table which shows the proliferative score a
display(
    eda_outputs["merged_scores"][
        ["proliferative_score", "apoptosis_resistance_score", "stage_simple", "stag
    ].head(10)
)
```

7. Inspect the PCA coordinates used for dimensionality reduction

This table contains the PCA coordinates for each COAD sample after dimensionality reduction of the 40-gene hallmark matrix. PCA allows us to visualize the dominant axes of

variation in the selected gene set and check whether samples appear to separate by tumor stage or by other structure present in the data.

In [ ]:
```
# Finally, we can also display the PCA table which shows the first 10 rows of the P
display(eda_outputs["pca_df"].head(10))
```

8. Inspect the K-means cluster evaluation table

We used K-means clustering as an exploratory grouping method and compared candidate values of k using silhouette score. This table summarizes the clustering results for each tested value of k, including inertia and silhouette score, and helps identify the most reasonable first-pass number of clusters for the COAD hallmark-expression data.

In [ ]:
```
# We can also display the cluster-evaluation table, which shows the inertia and sil
display(eda_outputs["cluster_eval"])
```

9. Review the saved output figures

These figures summarize our current exploratory analysis. The hallmark-score scatterplot compares the two hallmark programs directly across samples, the stage-based boxplot asks whether those scores vary across simplified tumor stages, and the PCA and clustering plots visualize broader structure within the 40-gene hallmark panel. The silhouette plot helps justify the current clustering choice by showing how cluster quality changes with the number of clusters.

In [ ]:
```
# Finally, we can also display the major figures we generated during our EDA workfl
from IPython.display import Image, display

results_dir = PROJECT_ROOT / "results"

display(Image(filename=str(results_dir / "COAD_hallmark_score_scatter.png")))
display(Image(filename=str(results_dir / "COAD_hallmark_scores_by_stage.png")))
display(Image(filename=str(results_dir / "COAD_PCA_by_stage.png")))
display(Image(filename=str(results_dir / "COAD_PCA_by_cluster.png")))
display(Image(filename=str(results_dir / "COAD_kmeans_silhouette.png")))
```

10. Display the UMAP coordinates

UMAP provides a nonlinear view of local structure in the same hallmark-gene matrix. We use it here as a complementary visualization to PCA rather than as a final conclusion. Comparing PCA and UMAP helps us judge whether the apparent structure is stable across different dimensionality-reduction approaches.

In [ ]:
```
# If UMAP was available and dovnloaded in our current environment, we can also disp
if eda_outputs["umap_df"] is not None:
    display(eda_outputs["umap_df"].head(10))
    display(Image(filename=str(results_dir / "COAD_UMAP_by_stage.png")))
    display(Image(filename=str(results_dir / "COAD_UMAP_by_cluster.png")))
```

```
else:
    print("No UMAP available.")
```

Overall Unsupervised Patterns Interpretation

Our unsupervised results suggest that the two hallmark programs are related but do not clearly separate tumor stage. In our hallmark-score scatterplot, samples with **higher proliferative scores** generally also tended to have **higher apoptosis-resistance scores**, indicating partial coordination between the two hallmarks. However, the stage-based boxplots clearly show substantial overlap across Stage I–IV, suggesting that these summary scores alone **do not cleanly distinguish stage**. On a similar note, PCA and UMAP did not reveal strong stage-separated groupings, and the best K-means solution had a low silhouette score of 0.154 for k = 3, indicating there was a weak cluster structure in our 40-gene feature space. Because the PCA, UMAP, and clustering analyses showed only weak separation by stage, we expected the supervised classifier to capture at most modest stage-related signal from this 40-gene hallmark panel.

11. Run the supervised training/validation workflow

In this step, we transition from exploratory analysis to supervised learning. Our goal is now to predict whether a COAD tumor belongs to the **Early** or **Late** stage group using hallmark-expression features. Following the workflow discussed in class, we use the **training set** to fit candidate models, the **validation set** to compare and select the final model, and the **test set** only once at the end to estimate final out-of-sample performance.

In [ ]:
```
# After completing the EDA, we now move to supervised learning.
# This fits candidate models on the training set, compares them on the validation s
# selects the final model, and then evaluates that selected model once on the held-
supervised_results = main.run_supervised_modeling()
```

12. Compare baseline and improved model metrics

This table compares the baseline and improved logistic-regression models on both the training and validation datasets. Reporting both in-sample and out-of-sample metrics helps us determine whether the model is learning a useful biological pattern that generalizes, rather than simply memorizing the training data.

In [ ]:
```
# We can also display the ROC curve comparison data for the validation set
# compares the baseline and improved models.
display(supervised_results["comparison_df"])
```

13. Display the validation ROC comparison

The ROC curves summarize how well each model separates Early-stage from Late-stage tumors across all possible decision thresholds. Because logistic regression outputs

probability scores, AUROC is a useful threshold-independent metric for comparing the baseline and improved models.

In [ ]:
```
# Finally, we can also display the ROC curve comparison figure for the validation s
display(Image(filename=str(results_dir / "COAD_validation_ROC_comparison.png")))
```

14. Display the validation confusion matrices

The confusion matrices show model behavior at the default decision threshold of 0.5. They make it easier to see the balance between correctly identifying Late-stage tumors and incorrectly labeling Early-stage tumors as Late, which is important for interpreting model error in a clinically meaningful way.

In [ ]:
```
# We can also display the confusion matrix comparison figure for the validation set
display(Image(filename=str(results_dir / "COAD_validation_confusion_matrices.png")))
```

15. Top coefficients from the improved model

These coefficients provide a first-pass look at which features most strongly influence the improved logistic-regression model. Because logistic regression is interpretable, the coefficient table helps connect predictive performance back to the biological story of proliferative signaling, apoptosis resistance, and tumor progression.

In [ ]:
```
#  display the coefficient table for the improved model which shows the top feature
display(supervised_results["improved"]["coef_df"].head(15))
```

Supervised Results Interpretation

Our supervised results show that our **improved** logistic-regression model generalized slightly better than our **baseline** model, but performance remained modest overall. On the validation set, the **improved** model achieved **higher AUROC**, balanced accuracy, and accuracy than the **baseline**, which suggests to us that adding hallmark-summary scores, age, class balancing, and stronger regularization was directionally helpful. However, the gap between training and validation performance remained substantial for both of our models, which indicates that the selected hallmark-expression features alone do not provide strong stage-prediction power in our dataset. Because the baseline model performed much better on the training set than on the validation set, we interpreted this pattern as overfitting and designed the improved model to reduce variance and improve generalization.

16. Evaluate the selected final model on the held-out test set

Following the final workflow discussed in class, we used the validation set to compare the baseline and improved models and then applied only the selected final model to the unseen test set. This provides our final estimate of out-of-sample performance on COAD samples that were not used during training or model selection.

In [ ]:
```
# display the final test results for the selected model on the held-out test set, i
display(supervised_results["final_test_results"]["test_metrics"])
print("Selected model from validation:", supervised_results["selected_model_name"])

# ROC curve and confusion matrix for the final selected model on the held-out test
display(Image(filename=str(results_dir / "COAD_final_test_ROC.png")))
display(Image(filename=str(results_dir / "COAD_final_test_confusion_matrix.png")))
display(supervised_results["final_metrics_df"])
```

Final Test-Set Interpretation

The held-out test set provides our strongest estimate of model generalization because it was not used to fit the model or to choose between candidate models. If the test-set metrics are similar to the validation-set metrics, that suggests our validation estimate was reasonably stable. If the test performance drops noticeably, that would suggest that the model still has limited robustness and remains somewhat sensitive to the specific training/validation split.

In our case, the final selected model achieved **0.589 AUROC**, **0.549 accuracy**, and **0.547 balanced accuracy** on the test set. These values were slightly lower than the validation-set metrics, suggesting that the model's generalization remained limited when applied to completely unseen data. Overall, the test results indicate that our final model provides only **modest** discrimination between Early- and Late-stage COAD tumors and is not strong enough to support any clinical predictive use.

Verify and validate your analysis:

We evaluated model performance using both **threshold-independent** and **threshold-dependent** metrics. **AUROC** was our primary threshold-independent metric because logistic regression outputs probabilities, and AUROC summarizes how well the model ranks Late-stage tumors above Early-stage tumors across all thresholds. At the default threshold of **0.5**, we also examined the **confusion matrix**, **accuracy**, **balanced accuracy**, **precision**, **recall**, and **F1 score**.

We first used the **training set** to fit our models and estimate in-sample performance. We then used the **validation set** to compare candidate models and assess out-of-sample generalization. The baseline logistic-regression model performed very well on the training set (**AUROC = 0.974**, **accuracy = 0.900**) but much worse on the validation set (**AUROC = 0.608**, **accuracy = 0.583**), which suggests overfitting. The improved model reduced training-set performance somewhat (**AUROC = 0.917**, **accuracy = 0.857**) but modestly improved validation-set performance (**AUROC = 0.637**, **accuracy = 0.597**, **balanced accuracy = 0.605**), indicating that adding hallmark-summary scores, age at diagnosis, class balancing, and stronger regularization reduced overfitting slightly.

Following the final workflow from class, we then selected the better-performing model based on **validation AUROC** and applied that selected model once to the **held-out test set**.

This test-set evaluation provides our final out-of-sample error estimate because the test data were not used during model fitting or model selection. Our final selected model achieved **0.589 AUROC**, **0.549 accuracy**, and **0.547 balanced accuracy** on the test set. The test confusion matrix showed 22 correctly classified Early-stage tumors and 17 correctly classified Late-stage tumors, with 16 misclassifications in each error direction, further emphasizing that the classifier's discrimination was limited.

Overall, our model shows **modest predictive ability** rather than strong stage discrimination. The gap between training performance and validation/test performance indicates that the selected hallmark-expression features capture some biologically meaningful signal related to stage, but not enough to produce a robust stage classifier on their own.

Conclusions and Ethical Implications:

Conclusions

Our final analysis suggests that our two selected hallmark programs are biologically relevant in COAD, but they do not strongly track tumor stage on their own. In the unsupervised analysis, proliferative-signaling scores and apoptosis-resistance scores showed some positive association across samples, but our stage-based boxplots overlapped substantially. PCA and UMAP also did not show strong separation of Stage I–IV tumors, and the best K-means solution had a low silhouette score of **0.154**. This indicated to us that there was a weak cluster structure in our 40-gene feature space.

In the supervised analysis, the improved logistic-regression model performed better than the baseline on the validation set and was therefore selected as the final model. When we applied this selected model to the held-out test set, its performance remained modest (**test AUROC = 0.589**, **test balanced accuracy = 0.547**). Because the test set was not used during model fitting or model selection, these values provide our final estimate of out-of-sample performance. Taken together, the validation and test results suggest that the two hallmark programs contain some stage-related information, but that signal is weak and not sufficient by itself to strongly predict simplified tumor stage in this dataset.

From a modeling perspective, our results highlight a key lesson: biological relevance does not guarantee predictive power. While the selected hallmark genes are mechanistically meaningful, they capture only a subset of the complexity driving tumor progression. Tumor stage reflects not only intrinsic tumor biology but also interactions with the tumor microenvironment, immune system, and broader systemic factors that are not captured in a 40-gene panel. Our findings are also consistent with the broader cancer-biology literature in suggesting that hallmark-associated expression programs are biologically informative, but they are not by themselves sufficient to fully explain a complex clinical phenotype such as tumor stage.

Ethical Implications

In an ethical sense, these findings carry important ethical considerations for the use of machine learning in biomedical contexts. First, they emphasize that models built on limited datasets and showing only modest held-out test performance should not be used for clinical decision-making without extensive external validation. Misclassification of tumor stage (particularly underestimating late-stage disease) could lead to inappropriate treatment decisions and harm to patients.

Second, our dataset represents a subsampled and potentially non-representative cohort, meaning that learned patterns may not generalize across diverse patient populations. If such models were deployed without addressing these biases, they could contribute to health disparities, especially for underrepresented groups in genomic datasets.

Finally, our work highlights the importance of interpretability and transparency. While more complex models could potentially improve predictive performance, simpler models like logistic regression allow us to understand which biological features are contributing to predictions. In a clinical setting, this interpretability is critical for building trust and ensuring that models are used responsibly.

Overall, our study supports the idea that hallmark-based gene expression analysis is valuable for understanding tumor biology, but insufficient on its own for clinical prediction.

Limitations and Future Work:

Limitations

Several limitations in our study likely contributed to the modest predictive performance and should be considered when interpreting our results:

1. Our analysis focused on a 40-gene hallmark panel, which, while biologically interpretable, captures only a small fraction of the molecular complexity of COAD. Important pathways such as immune response, angiogenesis, metabolism, and genome instability were not included.
2. The training dataset contains a relatively small number of COAD samples which limits statistical power and increases the risk of overfitting. Additionally, variability in metadata completeness and tumor annotation introduces noise into the analysis.
3. We reduced tumor stage into a binary classification (Early vs. Late), which may oversimplify meaningful biological differences between stages. This likely reduces the model's ability to detect more nuanced relationships.
4. Our evaluation does not fully test generalizability to independent datasets or clinical populations.
5. Our model relies solely on RNA-expression data, ignoring other important data types such as DNA mutations, copy-number variation, epigenetics, and proteomics, all of which contribute to tumor behavior.

Future Work

To build on our findings, several improvements could strengthen both the biological insight and predictive performance of this analysis:

1. Incorporating additional cancer hallmarks (e.g., angiogenesis, immune evasion, genome instability) could provide a more complete representation of tumor biology and improve model performance.
2. Instead of relying on individual gene expression, applying methods such as Gene Set Variation Analysis or enrichment-based scoring could better capture coordinated pathway activity.
3. Combining RNA expression with mutation data (e.g., KRAS, TP53 status), copy-number alterations, or methylation profiles would allow for a more comprehensive model of tumor progression.
4. Predicting full tumor stage (I–IV) or clinically relevant outcomes such as survival or recurrence may provide a more meaningful assessment of biological relevance.
5. Expanding to larger datasets or combining multiple cohorts would improve generalizability and reduce bias, especially across different demographic and clinical populations.
6. Testing the model on independent datasets is critical before any real-world application.

Notes:

- **Completed:**
  - Data and Disease Background: Dani and Will
  - Data Analysis: Dani and Will
  - Verification/Validation: Dani and Will
  - Conclusions/Ethics: Will
  - Limitations/Future Work: Dani
- **To Do:**
  - Nothing at this time.
- **Questions/Concerns:**
  - Nothing at this time.
- **References:**
  - [1] S. Groves and S. Peirce-Cottler, “Cancer Hallmarks ABM – Quick Reference,” course handout, Module 4, Computational Biomedical BME 2315, University of Virginia, 2026.

- [2] M. Rahman et al., “Alternative preprocessing of RNA-Sequencing data in The Cancer Genome Atlas leads to improved analysis results,” Bioinformatics, vol. 31, no. 22, pp. 3666–3672, 2015, doi: 10.1093/bioinformatics/btv377.
- [3] National Center for Biotechnology Information, “GSE62944 - GEO Accession viewer,” Gene Expression Omnibus. [Online]. Available: https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE62944.
- [4] National Cancer Institute, “Colon Cancer Treatment (PDQ®)–Patient Version.” [Online]. Available: https://www.cancer.gov/types/colorectal/patient/colon-treatment-pdq
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- **AI Statement:**
  - External use of LLMs was utilized on a case-by-case basis, often for initial source discovery + inspiration and code debugging. These resources were very great for finding research done on particular Hallmarks and gene sets. GitHub Copilot was especially helpful with providing 'Inline Suggestions' for shortening time with boilerplate code and repetitive function naming. This was particularly helpful for implementing LaTeX format into our markdown segments to that we can easily see the equations being used in a clear sense. However, the final code structure, modeling choices, and interpretations were based on the course lectures, GitHub materials, and provided resources, with all final analysis completed by our team.

QUESTIONS FOR YOUR TA:

We have no questions at this time.